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PAPER

First-principles and experimental investigation of d^0 magnetism in proton-implanted HOPG

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Supplementary material for this article is available online

Abstract

In this study, we report a combined experimental and first-principles investigation of the emergence of room temperature ferromagnetism in Highly Oriented Pyrolytic Graphite (HOPG) irradiated with 1.2 MeV proton (H+) ion at fluences ranging from 10–85 pC μm^{-2}. Vibrating Sample Magnetometry reveals the enhancement of ferromagnetic signal at the ion fluence of 35 pC μm^{-2} with saturation magnetization 0.89 emu g^{-1}. X-ray diffraction and Raman spectroscopy reveal that the ion irradiation in implantation regime induced lattice disorder, reduced crystallite size and origin of D-band at 1347.84 cm^{-1}, indicating disorder in the sp^2 network at the ion fluence of 35 pC μm^{-2}. SRIM simulations reveal the formation of maximum carbon vacancy near the proton stopping range and the decline in the inter-vacancy distance with increasing fluence. First-principles calculations, utilizing spin-polarized DFT, indicates that a single monovacancy (V_{C1}) creates no net magnetic ordering in HOPG system, while hydrogen-passivated vacancies (H_{C1}) produce a significant local magnetic moment (2.9 μ_B). Divacancies ($V_{C1\text{-}C2}$ & $V_{C1\text{-}C3}$) stabilize ferromagnetism through exchange coupling with magnetic moment of 2.10 μ_B to 2.13 μ_B. Hydrogen at interstitial sites (H_{int}) further induces less spin-polarization but contributed to interlayer expansion, which has been corroborated with the XRD/Raman analysis of 35 pC μm^{-2} irradiated sample. This combined experimental and theoretical study provides microscopic understanding of defect induced d^0-magnetism in HOPG and suggests pathway for achieving lightweight magnetic materials for spintronic applications.

1. Introduction

Carbon, a p-block element, exhibits great potential in materials science and technology, due to its unique properties, arising from different allotropes (Q-Carbon [1], Graphite, CNT [2], Fullerenes [3], Diamond [4] etc.). The unique arrangement of σ- and π-bonds that binds the carbon atoms determines the properties and geometry of these carbon allotropes. The demand for lightweight magnetic materials has become pivotal in modern technology due to their utilization in applications such as spintronic, data storage, sensing, etc. The absence of d- or f- shell electrons in carbon-based allotropes makes the system magnetically neutral. The pursuit of inducing magnetism in intrinsically non-magnetic carbon-based materials, such as the Highly Oriented Pyrolytic Graphite (HOPG), has become a potential research interest in materials science and technology in recent decades, to provide lightweight magnetic materials for device applications [5–7]. The HOPG samples exhibit intrinsic diamagnetism due to delocalized π-electron system [8, 9]. Intrinsic and extrinsic defects have shown great potential to tailor electronic and magnetic properties of graphene [10]. The adsorption or doping of foreign magnetic atoms in non-magnetic lightweight materials significantly induces magnetic ordering but previous studies demonstrate that magnetism can also be achieved in HOPG, by introducing defects such as vacancies, interstitials by utilizing ion irradiation techniques [5, 11–15]. In addition to HOPG, several other systems have induced ferromagnetic ordering upon irradiation or implantation with non-magnetic ions (e.g., carbon, protons), suggesting that defects mediated magnetism (e.g., vacancy formation, strain effects, etc.) plays a significant role in these materials [16–18]. Esquinazi *et al.*, reported for the first time in 2003, that 2.25 MeV proton irradiation on HOPG can induce room temperature ferromagnetism [6]. It challenged the conventional perception of carbon as a non-magnetic element, sparking substantial research into the fundamental principles. To enhance the reproducibility of experimental findings by ion beam irradiation, a variety of experiments have been performed across the globe to address the onset of magnetism in carbon related structures, in pursuit of optimum energy, dose and sample orientation etc.

To unveil the mechanism behind the defect-induced magnetism in HOPG, the following studies have been performed theoretically and experimentally. Lehtinen *et al* [19] performed spin-polarized Density Functional Theory (DFT) calculations of vacancies and vacancy-hydrogen complexes in HOPG, which are most likely to appear under irradiation. They have pointed out that hydrogen capture at vacancy sites, induces two-fold magnetic ordering, as compared to that of the bare vacancy. Ohldag *et al* [20], utilizing X-ray magnetic circular dichroism, demonstrated that spin-polarization of the carbon π-electron system due to hydrogen dissociation and its bonding at atomic lattice defects created by proton ion irradiation, leads to ferromagnetic ordering in HOPG. Ion beam irradiation with low to moderate fluence enhances the magnetic ordering, while increasing irradiation fluence beyond the optimal limit, may cause the magnetic ordering to vanish due to defect saturation in the sample [21]. The structural analysis plays crucial role, to understand the relationship between defect and emerging magnetic ordering in HOPG post irradiation. Phillips *et al* [22] have utilized x-ray diffraction (XRD) technique to investigate the defects in graphite and found broadening and loss of intensity in graphite XRD peaks due to vacancy. Along with XRD, Raman spectroscopy has also been utilized to study the disorder in the graphite-based systems, which reveal valuable information about stacking of graphene layers and defects, such as vacancy defects, sp^3- type defect formation and lattice disorder in graphite-based systems [23].

The correlation of induced ferromagnetic ordering with the type of defect and the underlying mechanism behind them, can be established by the utilization of first-principles calculations such as, density functional theory along with experiments. Yazyev [24] has utilized mean-field Hubbard model and first-principles calculation to exhibit the role of single atom defect on ferromagnetic order in graphene-based materials. In his work, it has been concluded that hydrogen binding and interstitial-vacancy complexes play critical role in π-electron ferromagnetism of proton-bombarded graphite. In the present investigation, we have systematically performed the experiment with proton ion irradiation in implantation regime on HOPG samples, with varying ion fluence at 1.2 MeV. The objective of this study is: (1) To find out the optimal ion irradiation dose to induce ferromagnetic ordering in HOPG, (2) To correlate the structural changes with induced magnetic ordering, and (3) To uncover the fundamental mechanism behind defect induced ferromagnetic ordering in HOPG utilizing first-principles calculations.

2. Methods

2.1. Experimental methods

The highgrade HOPG samples having line width shorter than 1°, has been procured from SPI Supplies for experiments. The samples has been prepared by exfoliating 2 mm $\times$ 3 mm HOPG flakes, with a thickness of approximately 0.1 mm. The samples have been adhered to a Quartz/Si substrate utilizing GE varnish to perform the irradiation experiment.

The ion irradiation experiment has been performed at the Tandetron accelerator at the IIT Kanpur. The schematic of experimental setup utilized in the present experiment has been described in figure 1. The Tandetron accelerator consists of two consecutive linear accelerator columns, working in tandem with the same terminal voltage. A hydrogen plasma source has been utilized to generate negative hydrogen ions. An injection magnet has been utilized to inject the negative hydrogen ion into first stage as denoted by blue arrow in figure 1. The positive high voltage (HV) terminal attracts the negative ion, and accelerates them in the first stage. In the second stage, nitrogen gas has been utilized to strip electrons from the ion, based on experiment requirement, and convert the negative ion into positive ion by stripping electrons. The positive ion then experiences a repulsive force and gets accelerated away in tandem from the HV terminal.

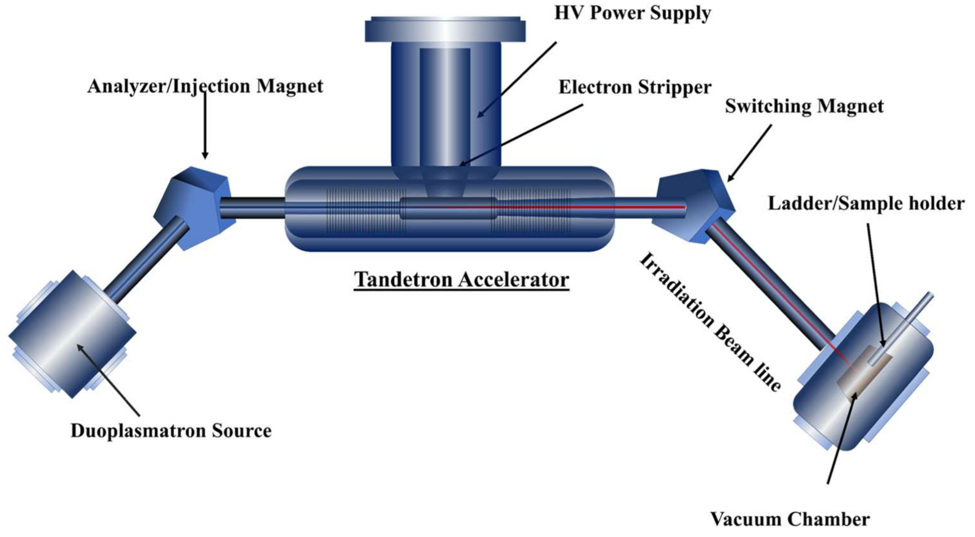

Figure 1. Schematic diagram of proton irradiation experiment setup utilizing Tandetron accelerator.

The irradiation of H^+ ions on HOPG samples have been performed by maintaining the current intensity around 10–20 nA mm^{-2}, which prevents annealing effects of the ion beam on the sample. An accurate beam current has been estimated by applying the positive electric field to the ladder, which helps to suppress the secondary electrons released from the HOPG sample. The ion fluences ranging from 10–85 pc μm^{-2} have been chosen for the irradiation.

The x-ray diffraction (XRD) measurement has been done utilizing Panalytical intelligent diffractometer (Model: Emp3) with Cu-K$_\alpha$ (1.5406 Å) source, at UGC-DAE-CSR Indore. The XRD measurement has been performed with step size 0.02° in the angle range of ($20° \leqslant \theta \leqslant 90°$). Raman Spectra have been obtained utilizing the excitation from 514.5 nm Ar^+ ion laser with a micro-Raman Spectrometer (Renishaw, UK, model in Via) in back scattering geometry. Raman spectroscopy has been performed to investigate chemical changes and detect the defect-related peaks induced by ion irradiation or implantation in the samples. The magnetic measurements have been performed utilizing vibrating sample magnetometer (VSM), at 'Advanced Center for Material Sciences (ACMS), IIT Kanpur'. The magnetic field have been varied in the range of ±18 kG. The samples have been mounted in a nonmagnetic holder with the help of pure Teflon tape.

2.2. Computational methods

Density Functional Theory (DFT) simulation has been performed to study pristine and defected graphite, utilizing an open source computational package Quantum Espresso [25]. Generalized Gradient Approximation (GGA) with (PBE) [26] exchange functional, has been utilized to account the interaction between electrons. Ultrasoft pseudopotential has been considered for ion core interaction. In order to consider the interaction between graphene layers, van der Waals (vdW) interaction, all the calculations have been performed with Grimme DFT+D3 correction. To study the contribution of spin-up and spin-down electrons on magnetic properties, we have performed spin-polarized calculation. The plane wave basis set with the cutoff energy of 70 Ry for the wavefunction and 560 Ry for the charge density, have been utilized for the calculation. A $4 \times 4 \times 4$ supercell containing 128 atoms, have been considered for inducing defects such as vacancy, hydrogen incorporation etc. This size of structure provides sufficient separation between defects to avoid spurious defect-defect interactions, while maintaining computational feasibility. A dense k-point mesh of $12 \times 12 \times 12$ Monkhorst–Pack [27] with total energy convergence threshold of 10^{-9} Ry has been set for self-consistent calculations. Structural optimization of all the samples has been executed, until the force on each atom has been found to be less than 10^{-5} Ry Bohr^{-1}. A k-point mesh of $6 \times 6 \times 6$ Monkhorst–Pack [27], has been utilized for the structural optimization. The structural and spin distribution plots have been visualized utilizing VESTA [28] and XCrySDen [29] packages. The stopping and range of ions in matter software (SRIM/TRIM) [30] has been utilized to estimate the implanted ion energy loss, defect density, penetration depth etc of MeV- ranged protons within the HOPG sample.

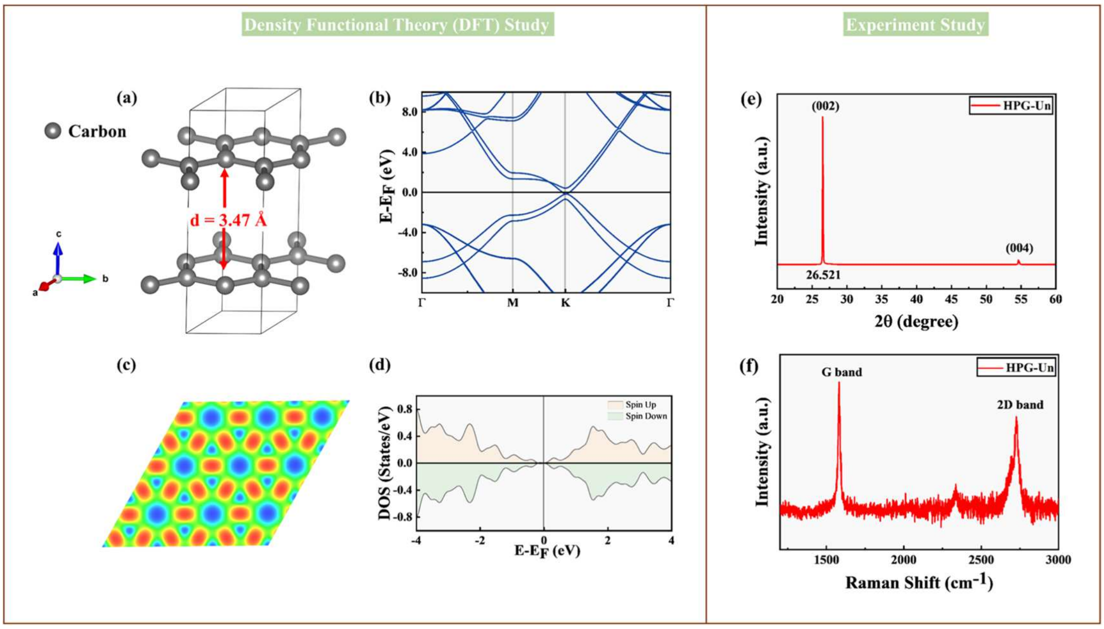

Figure 2. (a) Side view of theoretically optimized HOPG structure, (b) Band structure of HOPG, (c) Top view of Electron localized function (ELF) two-dimensional isosurface of HOPG, and (d) Density of States (DOS), of HOPG structure. (e) X-Ray Diffraction pattern and (f) Raman spectra, for un-irradiated HOPG (HPG-Un) sample.

3. Results

3.1. Structural, electronic and magnetic property of un-irradiated HOPG

Figure 2(a) exhibits the side view of Graphite, a layered allotrope of carbon, having the space group $P6_3/mmc$. The lattice parameters of optimized structure are a = 2.46 Å, b = 2.46 Å, c = 6.94 Å, which nicely matches with the previously reported values [31, 32]. Figure 2(b) depicts the band structure of graphite, which exhibits graphite's semi metallic characteristics. The Dirac-cone like dispersion exhibited by graphene is altered in graphite, due to interlayer interaction, introducing small overlaps between conduction and valance bands at K-point. Along the Γ-K and M-K paths, the band curvature reveals the in-plane anisotropy. The conduction is dominated by the high in-plane mobility due to the strong covalent bonding. The electron localization function (ELF) two-dimensional isosurface displayed in figure 2(c), reveals the localization of electron between carbon atoms, which elucidates the strong in-plain σ-bonding. The corresponding Density of States (DOS) plot exhibited in figure 2(d), reveals the symmetric nature of DOS for spin-up and spin-down states, which indicates the non-magnetic nature of graphite.

The X-ray diffraction (XRD) pattern shown in figure 2(e), elucidates the structural property of un-irradiated HOPG (HPG-Un). The XRD pattern of HPG-Un sample reveals a strong peak at 2θ position 26.521°, which nicely matches with graphite (002) peak. This Braggs equation (equation s1 in supplementary information) has been utilized to estimate the interlayer spacing 'd', which is found to be 3.354 Å for HPG-Un sample. This experimental value of d matches well, with the theoretically estimated (DFT analysis) value (3.47 Å) of interlayer spacing. Based on the analysis of (002) peak, the crystallite size 'D_c', calculated utilizing equation s2, has been estimated to be 697.927 Å, which sets the reference for crystallite order and defect density for irradiation experiment.

The Raman spectroscopy further provides the information of structural environment of unirradiated HOPG. The Raman spectra of HPG-Un, including G and 2D-band have been displayed in figure 2(f). The Raman spectra of HPG-Un displays a prominent G-band at 1582.02 cm^{-1}, corresponding to the double degenerated E_{2g} phonon mode associated with the in-plane vibrational motion of the sp^2-hybridized carbon atoms in hexagonal lattice arrangement. The 2D band at 2718.41 cm^{-1} is the second-order peak, which emerges due to the double resonant electronic process. The intensity ratio (I_{2D}/I_G) of 2D-band with respect to the D-band estimated from figure 2(b) is 0.614, which reflects strong interlayer coupling and highly organized stacking sequence. The absence of D-band a signature of sp^3 defects, highlights the defect free, highly ordered state of un-irradiated HOPG. The combined XRD and raman data, points that HPG-Un is highly crystalline and defect free, which provides an initial reference for further defect engineering and ion irradiation experiments.

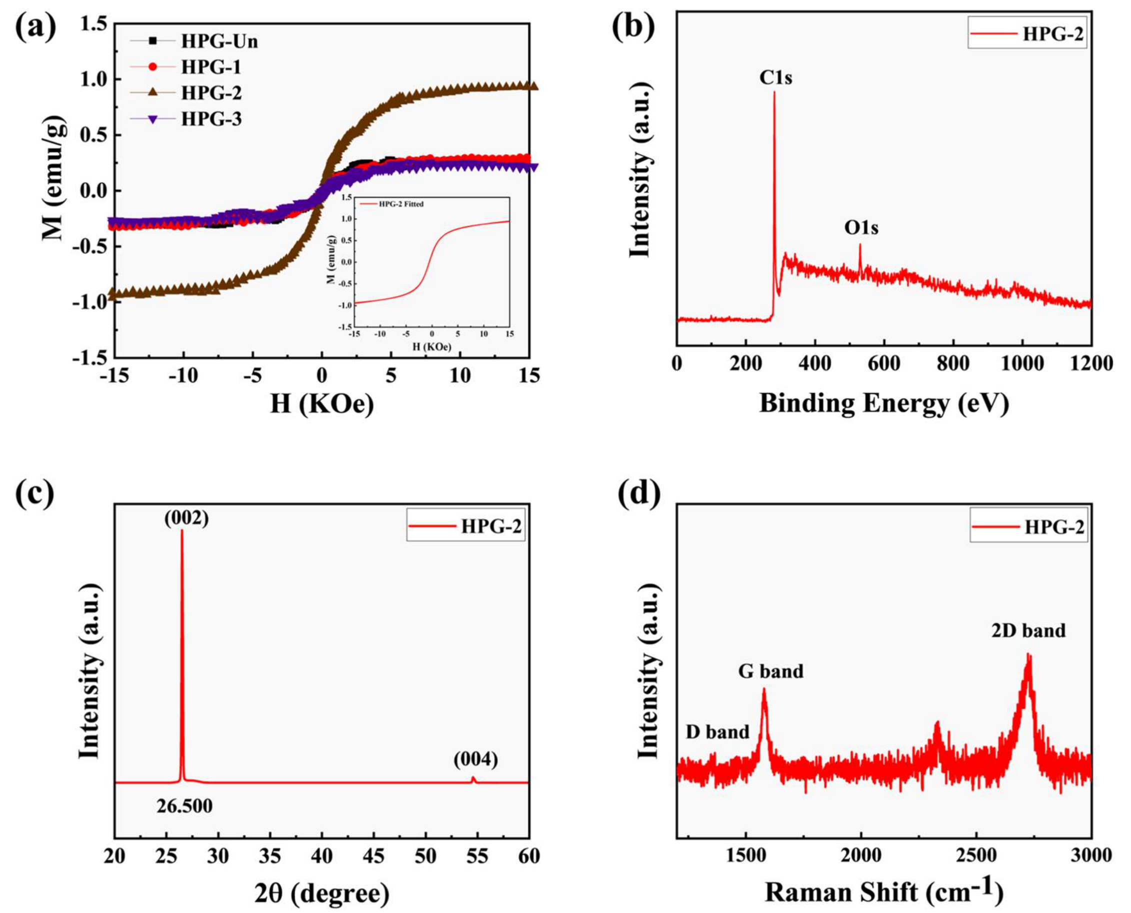

Figure 3. (a) M-H curve of unirradiated and irradiated samples, HPG-Un, HPG-1, HPG-2 and HPG-3 respectively, (b) XPS spectrum, (c) XRD pattern and (d) Raman spectrum for proton irradiated HOPG (HPG-2) sample.

3.2. Structural and magnetic property of irradiated HOPG

The 1.2 MeV proton irradiation causes significant structural changes in HOPG, which alters its magnetic properties. The M-H curves after subtracting the diamagnetic part for the samples HPG-Un, HPG-1 [10 pC μm^{-2}], HPG-2 [35 pC μm^{-2}], and HPG-3 [85 pC μm^{-2}], have been exhibited in figure 3(a). The magnetization behavior for sample HPG-1 and HPG-3 is nearly similar to HPG-Un sample. However, at an optimal fluence of 35 pC μm^{-2} (HPG-2), a significant enhancement in magnetization at room-temperature has been observed as compared to that for the unirradiated sample (HPG-Un). The quantification of the observed magnetization has been realized by utilizing the following equation [33, 34],

$$M(H) = \left[2\frac{M_S}{\pi}\tan^{-1}\left\{\frac{H \pm H_C}{H_c}\tan\left(\frac{\pi M_R}{2M_S}\right)\right\}\right] + \chi H \tag{1}$$

Where, 'M_R' is the remanence magnetization, 'M_S' is saturation magnetization, 'H_C' is the intrinsic coercivity and 'χ' is the magnetic susceptibility. The first part of the equation (3), represents the ferromagnetic hysteresis and the second part represents the possible paramagnetic contribution. The fitted curve of HPG-2 sample has been displayed in figure 3(a) in the inset. The extracted parameters M_S, M_R, and H_C from the fitted curve are 0.89 emu g^{-1}, 0.18 emu g^{-1}, and 522.8 Oe, respectively. (HPG-2) samples have been subjected to X-ray photoelectron spectroscopy (XPS) in order to rule out the potential of extrinsic impurity-induced magnetism. There have no discernible signs of Fe, Co, Ni, or other transition-metal impurities present in the XPS spectra of HPG-2 as shown in figure 3(b). XPS spectra of un-irradiated sample (HPG-Un) has also been added in supplementary information as Fig. S3 for reference. These findings point out that ferromagnetic ordering in HPG-2 sample is due to the ion-implantation and irradiation-induced defect and structural changes.

The XRD pattern of proton irradiated HPG-2 sample has been exhibited in figure 3(c), which reveals significant structural changes, which correlate with enhanced magnetization. The (002) peak, which reveals the information of interlayer spacing *d* in graphitic materials, exhibit a shifts from 26.521° (HPG-Un) to lower angle 26.501°(HPG-2) post irradiation. According to Bragg's law, this angular reduction corresponds to a minor yet significant ($\Delta d \approx 0.002$ Å) expansion of the interlayer distance from 3.354 Å to 3.356 Å. This shift

Table 1. The peak position and crystallite size extracted from (002) peak of XRD Pattern of un-irradiated and irradiated HOPG samples.

Sample Name	Fluence (pc/um^2)	Peak position $2\theta_{(002)}$ (degree)	Crystallite size (Angstrom)
HPG-Un	—	26.521	697.927
HPG-1	10	26.521	662.150
HPG-2	35	26.501	616.671
HPG-3	85	26.521	684.473

Table 2. G-band, 2D-band and D-band position with I_D/I_G and I_{2D}/I_G ratios of un-irradiated and irradiated HOPG samples.

Sample name	G-band position (cm^{-1})	2D-band position (cm^{-1})	D-band position (cm^{-1})	I_D/I_G	I_{2D}/I_G
HPG-Un	1582.02	2718.41	—	—	0.614
HPG-1	1580.79	2717.52	—	—	0.640
HPG-2	1579.27	2714.85	1347.84	0.184	1.332
HPG-3	1581.13	2715.83	1359.33	0.206	0.644

likely arises from the possibility of hydrogen ion incorporation at interstitial sites between the graphene layers. Further, XRD analysis of HPG-2 sample, reveals the reduction in crystallite size from 697.927 Å (HPG-Un) to 616.671 Å (HPG-2). The reduction of crystallinity order in HPG-2 sample demonstrates increased lattice disorder and defect density such as vacancy, interstitials, due to proton implantation. The crystallite size and peak position for lower fluence 10 pC μm^{-2} (HPG-1) and higher fluence 85 pC μm^{-2} (HPG-3), have been estimated from figure S1 and the values have been depicted in table 1. Both the HPG-1 and HPG-3 samples exhibit (002) peaks at nearly identical angles, which matches well with the HPG-Un sample.

Figure 3(d) exhibits the Raman spectra for HPG-2 sample irradiated with fluence of 35 pC μm^{-2}. A tiny but significant peak of D-band has been detected at 1347.8 cm^{-1}. The origin of D-band, a disordered active mode, marks the presence of lattice defect such as vacancies, sp^3 hybridized defects in HOPG, post proton irradiation.

When the periodicity of carbon atoms in graphite is disturbed by proton irradiation, defects such as point defects, vacancies, introduce the additional D peak (1347.843 cm^{-1}) in the Raman spectrum of the disordered irradiated sample HPG-2. The G-band, 2D-band and D-band positions with I_D/I_G and I_{2D}/I_G for respective un-irradiated and irradiated samples, have been displayed in table 2. The Raman spectra of samples HPG-1 and HPG-3 have been added in the supplementary data as figure S2. The intensity ratio I_D/I_G exhibits the quantitative assessment of defect after irradiation. The I_D/I_G ratio is nearly zero for HPG-Un and HPG-1 samples, which confirms the presence of largely undisturbed sp^2 network in these HOPG samples. Further, the I_D/I_G ratio increases to 0.184 in HPG-2 sample, which depicts the moderate defect density within the graphene planes. The proton irradiation not only induces defects but also alters the long-range hexagonal lattice symmetry, but also influences the interlayer coupling. The enhancement in I_{2D}/I_G ratio from 0.614 to 1.332 has been observed in HPG-2 sample. Such an enhancement in I_{2D}/I_G ratio indicates weaker interlayer coupling and disturbed stacking order or trapping proton ion between layers. Furthermore, the Raman spectra of HPG-3 exhibits highest I_D/I_G ratio of 0.206 and red-shift in D-band, which indicates the increased defect density within the graphene plane. It shows non-monotonic dependence of the I_{2D}/I_G ratio on fluence. This behavior exhibits two different effect generated by proton irradiation. At fluence 35 pC μm^{-2}, the partial breaking of AB stacking and modest interlayer expansion occurs, while at higher fluence 85 pC μm^{-2}, sp^3-type defects increase.

3.3. Proton induced defect density and damage profile in HOPG

SRIM simulation based on Monte-Carlo technique, has been performed to gainthe insight on proton implanted HOPG samples. SRIM provides a detailed analysis of ion-matter interactions by calculating the ion penetration depth, defect formation and energy dissipation, within the target material. HOPG structure has been modeled by considering graphite allotrope with density of 2.253 g cm^{-3}, corresponding to atomic density of 1.129×10^{23} atoms cm^{-3}. SRIM estimates a nuclear and electronic energy loss of 0.2047 MeV $(mg\ cm^{-2})^{-1}$ and 1.294×10^{-4} MeV $(mg\ cm^{-2})^{-1}$, respectively for 1.2 MeV proton irradiation on HOPG. Nuclear energy loss dominates over electronic energy loss at 1.2 MeV proton energy, which favors vacancy-type defect formation. The 1.2 MeV proton penetrates up to a depth of approximately 16.43 μm in HOPG sample as predicted by SRIM. This penetration depth is well within the HOPG sample thickness (100 μm). Since the ion penetration depth is smaller than the sample thickness of the HOPG samples, the process falls within the ion implantation regime.

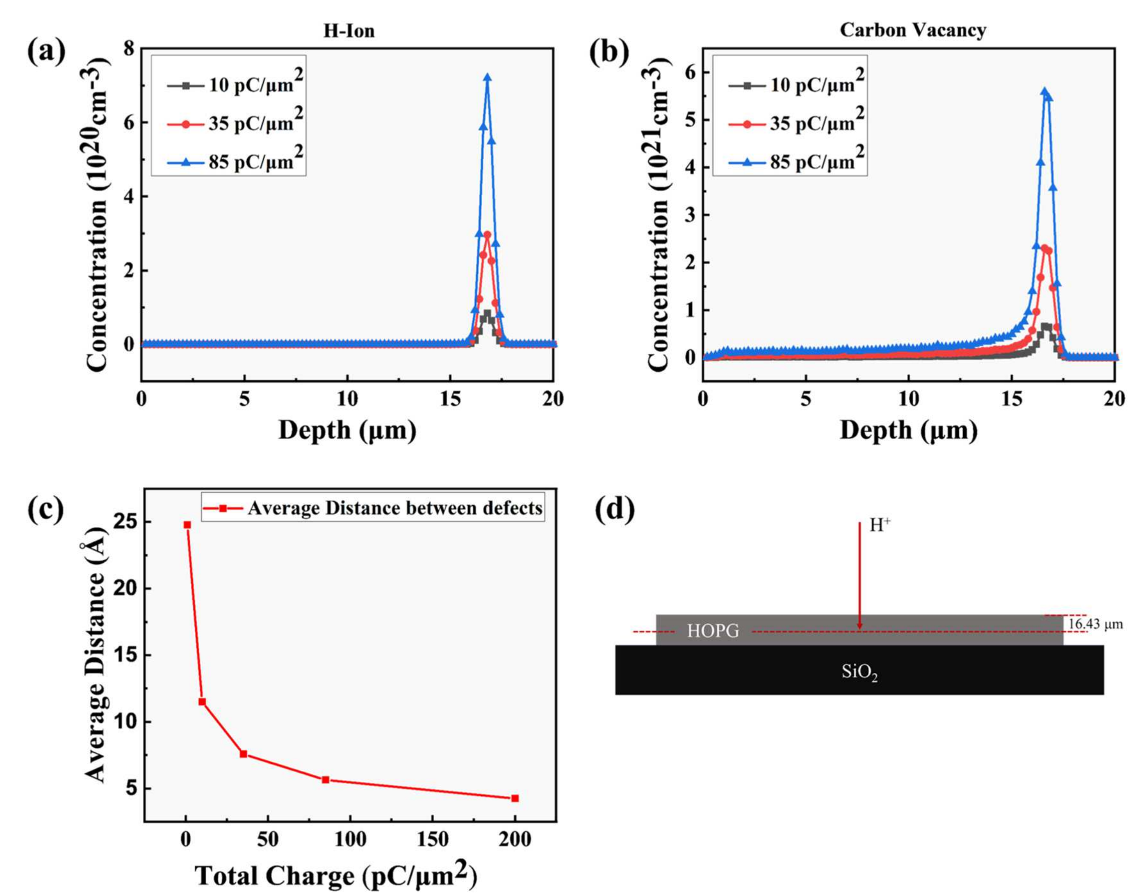

Figure 4. (a) Concentration of Proton (H^+) ion with respect to depth, (b) Concentration of Carbon vacancy with respect to depth, for ion fluence 10 pC μm^{-2}, 35 pC μm^{-2} and 85 pC μm^{-2}, (c) Average distance between vacancy defects in HOPG samples with respect to the proton fluence and (d) Schematic of proton (H^+) ion irradiation on HOPG sample.

Figure 4(a) displays the distribution of 1.2 MeV proton ion with respect to the depth of the HOPG. The large peak just before the end of the range, indicates a behavior similar to a Bragg peak, in which the protons deposit the majority of their energy just before they come to rest. The concentration of proton ion reaches 10^{19}–10^{20} cm^{-3}, depending on the fluence. This shows the energy dissipation characteristic of the proton ion within the HOPG target material is in line with ion implantation technique. It has been observed that each incident proton produces approximately 15 vacancies in HOPG, which enables better control on the fluence dependent defect formation in the HOPG. The depth dependence of carbon vacancy concentration for different proton ion fluences, have been displayed in figure 4(b). The maximum vacancy concentration $\sim 10^{21}$ cm^{-3} has been observed near the stopping range of 16.43 μm for 1.2 MeV proton in HOPG. The increment in peak intensity with ion fluence indicates that higher vacancy concentration is achieved for higher proton fluence. The further increment in the ion fluence may lead to the excessive lattice damage in the sample, which may reduce the distance between defects comparable to the lattice parameter, leading to the amorphization of the sample. Figure 4(c) exhibits the average distance between vacancy defects, with respect to ion fluences. The ion fluence range (10–85) pC μm^{-2} has been selected to vary the defect density. Increasing the ion fluence results in a decrease defect-defect distance from 1.15 nm to 0.75 nm for 10 pC μm^{-2} and 35 pC μm^{-2} respectively. The average distance between vacancies continuously decreases as we increase the ion fluence within the defect solubility limit, which is not the case in the present experiment.

3.4. Density functional theory analysis

The defect correlation with induced magnetism is a multifaceted process, which requires careful analysis of defects created and their theoretical estimation as per sample requirements. The first-principles calculation utilizing DFT simulation have been performed to identify the contribution of possible defects e.g. carbon monovacancy (V_{C1}), hydrogen at carbon vacancy site (H_{C1}), hydrogen at interstitial site (H_{int}) and the effect of interaction between two carbon vacancies ($V_{C1\text{-}C2}$ and $V_{C1\text{-}C3}$) on the magnetic ordering. A systematic study has been performed to quantify the contribution of defects on the magnetic ordering. Table 3 exhibits the total

Table 3. The total magnetic moment (M_{total}) and interlayer spacing (d) of pristine and defect incorporated HOPG samples.

Defect Type	d (Å)	M_{total} (μ_B)
pristine	3.47	0.00
V_{C1}	3.46	0.00
H_{C1}	3.46	2.87
$V_{C1\text{-}C2}$	3.47	2.13
$V_{C1\text{-}C3}$	3.46	2.10
H_{int}	3.60	0.02

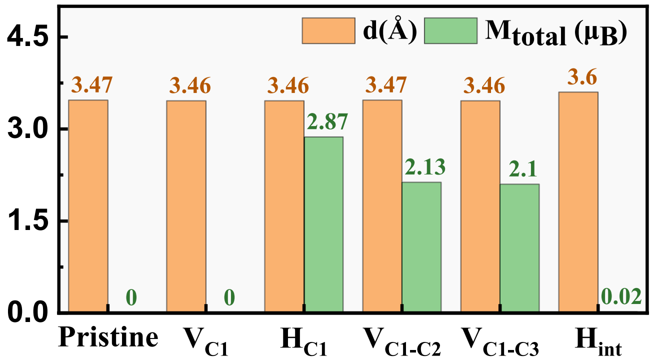

Figure 5. The graphical representation of interlayer spacing (d) and magnetic moment (M_{total}) of pristine and defect incorporated HOPG samples.

magnetic moment and variation in interlayer distance with respect to defect type. A graphical representation of this variation has been displayed in figure 5.

It has been observed that mono-vacancy (V_{C1}) does not produce net magnetization, the absence of net magnetization in V_{C1} arises from spin compensation due to reconstruction and interlayer interaction in multilayer HOPG, in contrast to isolated graphene monovacancies. The mono-vacancy filled by hydrogen atom (H_{C1}), gives rise to net magnetic ordering with a magnetic moment of approximately 2.87 μ_B of magnetic moment value. To account for the vacancy interactions; Carbon di-vacancy, $V_{C1\text{-}C2}$ and $V_{C1\text{-}C3}$, have been considered. The distances between vacancies C_1 and C_2 and that for C_1 and C_3 have been chosen to be 4.26 Å and 5.68 Å, respectively. Both these di-vacancy structures, $V_{C1\text{-}C2}$ and $V_{C1\text{-}C3}$, exhibit magnetic ordering having magnetic moment values of 2.13 μ_B and 2.10 μ_B respectively. This variation in magnetic moment in both the di-vacancy systems is due to the the strength of magnetic interaction with respect to the distance. It has been observed that hydrogen at interstitial site also contributes to magnetization value of 0.02 μ_B. The incorporation of defects in layer L_3 and interstitial sites between layers L_2 and L_3 also affects the interlayer spacing. The maximum change in interlayer spacing has been observed for hydrogen at interstitial site.

The distribution of spin density, has been calculated to distinguish the spin distribution for respective defect type. The spin density difference ($\Delta\rho = \rho_{up} - \rho_{down}$) isosurface has been displayed in figure 6. The positive and negative spin density difference ($\Delta\rho$), responsible for spin-up and spin-down denoted regions, have been displayed with yellow and blue colors respectively. Amongst the layers L_1, L_2, L_3 and L_4 in graphite, the layer L_3 has been considered for introducing defects, the side view and top view of the structures having spin density difference isosurface in layer L_3, has been displayed in figures 6(a)–(e). The yellow and blue region corresponding to net spin-up and spin-down density distribution, has been observed around the mono-vacancy, which results in zero magnetic moment as shown in table 3. When the vacancy V_{C1} is stabilized by hydrogen atom, a positive spin density difference isosurface is observed all over the layer L_3. The hydrogen in carbon matrix helps to stabilize the long-range ferromagnetic ordering as exhibited in figure 6(b). It has been observed that the distance between vacancies also influences the spin distribution as exhibited in figures 6(c) and 6(d). The interaction of vacancies V_{C1} and V_{C2}, enhances majority spin (Spin-up) contribution, which develops the ferromagnetic ordering in the system and also mediates the formation of pentagon between vacancies, which leads to the disruption in hexagonal lattice matrix. Increasing the vacancy distance from 4.26 Å and 5.68 Å, also favors positive spin density difference distribution with identical spin distribution around the vacancy. The yellow spin region corresponds to positive spin density distribution in $V_{C1\text{-}C3}$ structure form

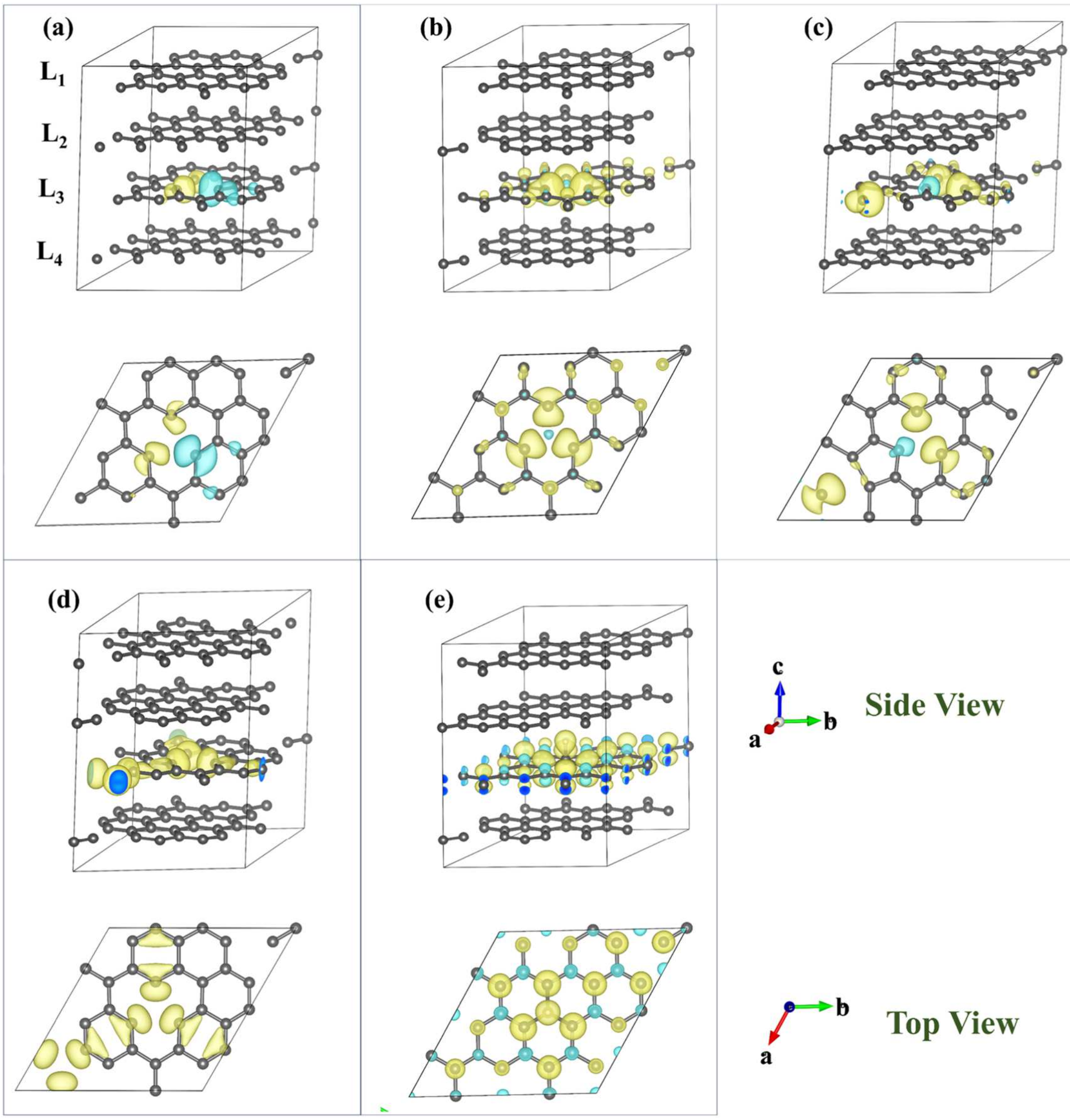

Figure 6. Top view and side view of spin density difference ($\rho_{up}-\rho_{down}$) isosurface for (a) Single mono-vacancy V_{C1}, (b) Carbon mono-vacancy filled by Hydrogen H_{C1}, (c) Carbon di-vacancy $V_{C1\text{-}C2}$ and (d) Carbon di-vacancy $V_{C1\text{-}C3}$ system (e) Hydrogen at interstitial site (H_{int}) of HOPG samples having isosurface value 0.0065. Yellow and blue regions represent the dominated majority and minority spin density distribution.

hexagonal type spin distribution, as displayed in figure S4. The hydrogen at interstitial site as displayed in figure 6(e), disrupts the spin distribution and this results in positive and negative spin density difference denoted by yellow and blue regions. The surrounding carbon atoms near hydrogen exhibit spin-up domination and second nearest neighbor carbon atom exhibits spin-down domination. This alternate spin-up and spin-down region is extended periodically over the lattice. The hydrogen at interstitial site disrupts the spin distribution for long range, but the presence of spin-up and spin-down spin regions, decreases the resultant magnetization.

The Total Density of States (DOS) and Partial Density of States (PDOS) further unveil the redistribution of electronic state for these systems. The changes in the electronic states after introducing defects, have been displayed in respective DOS and PDOS plots in figures 7(a)–(j)). The observed symmetry in spin-up and spin-down states for mono-vacancy V_{C1}, enunciates that the introduction of mono-vacancy changes the electronic states near the fermi level but still the spin-states follow the symmetric distributions as displayed in figure 7(a). This change in DOS primarily comes from the redistribution of electrons in 2p-orbitals carbon atom in Layer L_3 as displayed in figure 7(f). When the di-vacancy is introduced in the system, the occupancy of spin-up and spin-down electronic states reduces near the fermi level and gets distributed asymmetrically, as compared to mono-vacancy system as displayed in figures 7(c) and (d). This asymmetry between spin-up and spin-down states breaks the time reversal symmetry, which makes the system ferromagnetic having magnetic moment of 2.13 μ_B and 2.10 μ_B, for $V_{C1\text{-}C2}$ and $V_{C1\text{-}C3}$ systems, respectively. Similar behavior has also been observed in H_{C1} system. In this system hydrogen at vacancy site C1, alters the occupancy of spin-up and spin-down electronic

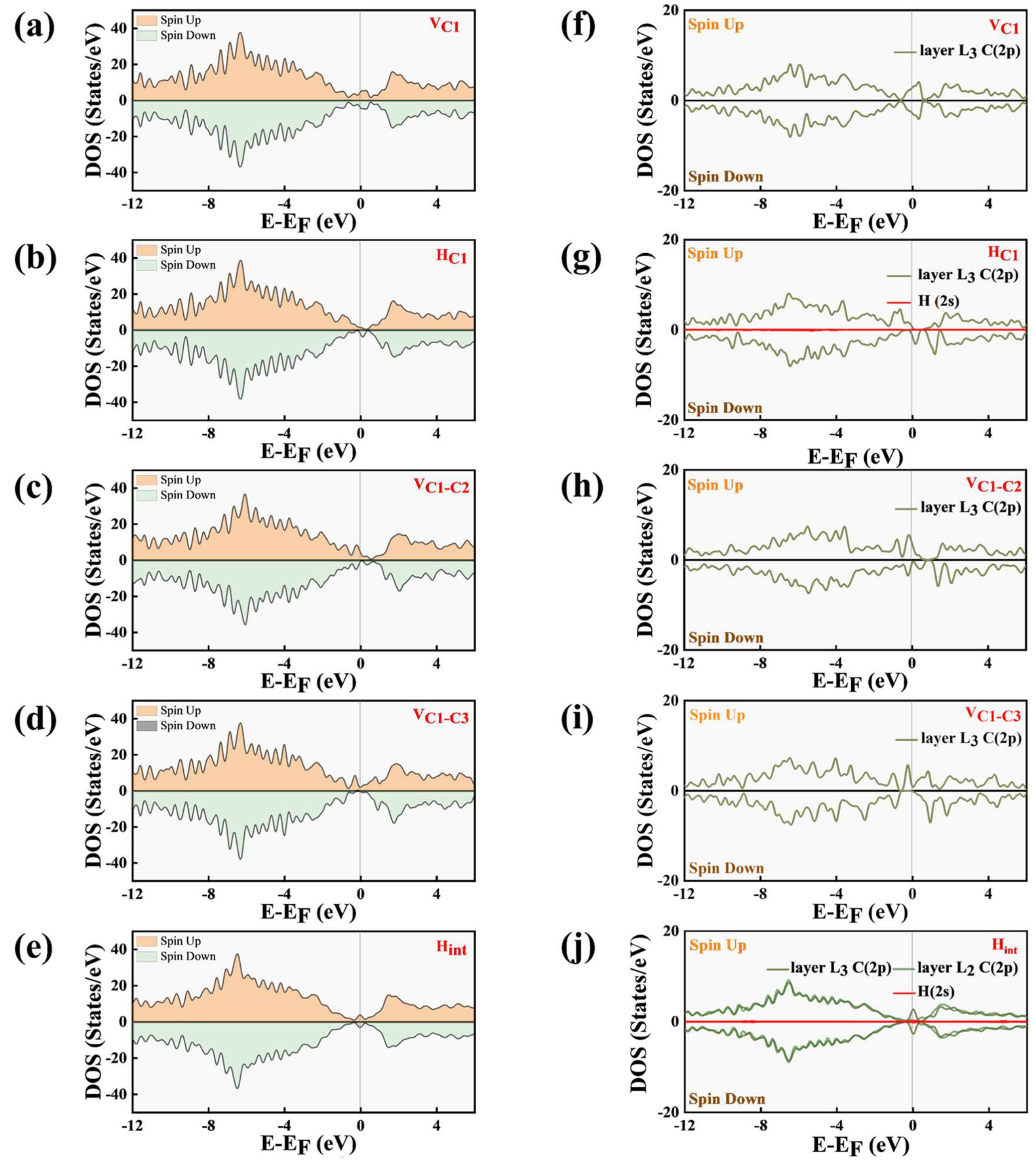

Figure 7. Total Density of States (DOS) for (a) Single mono-vacancy V_{C1}, (b) Carbon mono-vacancy fill by Hydrogen H_{C1}, (c) Carbon di-vacancy $V_{C1\text{-}C2}$, (d) Carbon di-vacancy $V_{C1\text{-}C3}$ and (e) Hydrogen at interstitial site (H_{int}) system and (f)(j) Partial Density of States (PDOS) for respective systems.

states of 2p-orbitals of carbon atoms of layer L_3, as shown in figure 7(g), causes significant asymmetry in spin-up and spin-down states. However, hydrogen at interstitial site alters the occupancy of electronic states for layer L_3. The significant occupancy of spin-up and spin-down state have been observed near fermi level, but the asymmetry in the spin-up and spin-down states is less as compared to other systems, figures 7(e) and (j).

To identify the dominant spin-states at fermi level and quantify their contribution, degree of spin-polarization has been calculated utilizing the following equation [35],

$$P = \frac{N_{\uparrow}(E_F) - N_{\downarrow}(E_F)}{N_{\uparrow}(E_F) + N_{\downarrow}(E_F)} \times 100\% \tag{2}$$

Where, $N_{\uparrow}(E_F)$ and $N_{\downarrow}(E_F)$ are the spin-up and spin-down density of states at fermi level respectively. The calculated value for system V_{C1}, H_{C1}, $V_{C1\text{-}C2}$, $V_{C1\text{-}C3}$ and H_{int}, have been displayed in table 4. $V_{C1\text{-}C2}$ system exhibits higher degree of spin polarization (56.65%), where the positive value represents the domination of spin-up at fermi level. The degree of spin-polarization reduces to 45.13%, when the distance between vacancies is increased ($V_{C1\text{-}C3}$), due to redistribution of spin-up and spin-down electronic states near fermi level. The V_{C1} system having mono-vacancy in the unit cell, exhibits zero degree of spin polarization due to the symmetrical

Table 4. Degree of spin polarization with spin-up and spin-down states at fermi level for V_{C1}-, H_{C1}-, $V_{C1\text{-}C2}$, $V_{C1\text{-}C3}$- and H_{int}- type structures.

Defect type	$N_{\uparrow}(E_F)$	$N_{\downarrow}(E_F)$	P (%)
$\mathbf{V_{C1}}$	4.08	4.08	0
$\mathbf{H_{C1}}$	1.15	2.08	−28.79
$\mathbf{V_{C1\text{-}C2}}$	4.59	1.27	56.65
$\mathbf{V_{C1\text{-}C3}}$	1.64	0.62	45.13
$\mathbf{H_{int}}$	3.55	3.53	0.28

spin-up and spin-down states. However, the degree of spin polarization is reversed to (−28.79%) for H_{C1} system, where the negative sign in degree of spin polarization indicates, that the fermi level has been majorly occupied by the spin-down states.

It has also been noticed that hydrogen at interstitial site (H_{int}) increases the occupancy of spin-up states $N_{\uparrow}(E_F)$ and spin-down states $N_{\downarrow}(E_F)$ at fermi level of approximately same order, which exhibits minor degree of spin-polarization. The changes in the system's magnetic properties, electronic states and the degree of spin polarizations after hydrogen interaction with carbon lattice, suggest that hydrogen not only creates optimal vacancy in the system but can also enhance the magnetic ordering by occupying the vacancy sites and interstitial sites.

4. Discussion

Proton irradiation in the implantationregime has been acknowledged as an excellent technique for creating controlled lattice defects in HOPG [36]. Previous experimental and theoretical findings suggest that defect induced quasi localized electronic states yields the magnetic ordering in HOPG [9, 20, 24, 37]. Extensive studies have been carried out on oxide materials such as ZnO to elucidate the mechanism of defect-induced magnetism. Mishra et al. in their work have shown that defects like Zn-interstitial and O-vacancy may lead to room temperature ferromagnetism in ZnO single crystal [38]. This can be understood by oxygen vacancy mediated bound magnetic polaron (BMP) model, where localized carriers mediate long-range magnetic coupling [39]. In the present work, weestablish an experimental and theoretical framework to elucidate the relationship between defect formation, hydrogen incorporation, and room-temperature ferromagnetism in HOPG irradiated by 1.2 MeV proton. The choice of 1.2 MeV H+ ion beam has been motivated by the suitability for achieving controlled generation of vacancy-type point defect within the implantation regime of HOPG. The point defect densities produced by proton beam implantation in HOPG is lower than that produced by heavy ions, which plays a crucial role in stabilizing magnetic ordering. J. Barzola-Quiquia *et al.* also have shown that Fe-ion produce large vacancy density and found out that a high vacancy density may prevent the existence of long range magnetic interaction through the graphene structure [40]. In order to identify the ideal ion dose to induce magnetism, the proton fluence range of 10–85 pC μm^{-2} was specifically chosen. This range was approximated by SRIM simulations performed for 1.2 MeV protons on HOPG, the simulations predict increasing vacancy density and decreasing average inter-vacancy distance with increasing fluence as exhibited in figure 4(c). A comparison of experiment procedure and findings of present work and previous reported work have been exhibited in table S1. The HPG-Un, HPG-1, and HPG-3 shows weak and nearly identical magnetic ordering. This magnetic ordering in initially presented in the HOPG samples due to intrinsic defects such grain boundaries, edges, and point defects [12], which are naturally presented in HOPG, since all the samples are exfoliated from a single HOPG sample. The magnetic characterization using VSM indicates that the ferromagnetic order enhances in HPG-2 sample, with a saturation magnetization of 0.89 emu g^{-1}. The structural analysis of HPG-2 sample reveals the generation of defect and interlayer decoupling at an optimal fluence of 35 pC μm^{-2}. XRD analysis of HPG-2 sample, reveals that the interlayer spacing gets enhanced from 3.354 Å to 3.356 Å and a significant reduction occurs in the crystallite size from 697.9 Å to 616.7 Å with respect to the unirradiated HPG-Un sample. Although, the observed change in the interlayer spacing in HPG-2 sample is minor and close to the experimental error margin in XRD, but its significance is reinforced by Raman spectroscopy. Raman spectroscopy further reveals that I_D/I_G ratio has increased to 0.184 due to presence of defect induced D-band in HPG-2 as displayed in figure 3(d). A significant increase in the I_{2D}/I_G ratio (1.332), after proton implantation, points towards the interlayer decoupling with substantial disordered stacking. This change in interlayer interaction could be due to hydrogen incorporation at the interstitial sites or the presence of vacancy type defects in the hexagonal lattice network of HOPG. At low fluence (10 pC μm^{-2}), defects are widely distributed with low defect density, the interaction between localized magnetic moments is insufficient to sustain long-range magnetic ordering. At an optimal fluence (35 pC μm^{-2}), the average distance between defects decreases to a critical range, enabling effective exchange interaction between localized moments, which leads to stabilization of ferromagnetic ordering. At higher fluence (85 pC μm^{-2}), excessive defect density introduces structural disorder, which reduces the net magnetization and exhibits similar behaviors as pristine HPG-Un sample [41]. Similarly, Yang *et al* have correlated defect density variation with altering mechanical properties of graphene utilizing MD simulations [42]. These findings exhibit the transformation of highly ordered sp^2 network to a partially disordered configuration with sp^3 bonding and altered π–π interactions, which establishes the condition for spin localization [9, 20].

To identify the contribution of defect type with emerging magnetic ordering, Spin-polarized DFT simulations have been performed by considering various defect configuration, such as V_{C1}, H_{C1}, $V_{C1\text{-}C2}$, $V_{C1\text{-}C3}$ and $H_{int.}$ These configurations highlight the magnetic behaviors and corresponding structural changes after introducing the defects. In pristine structure, the interlayer spacing has been found to be 3.47 Å (DFT) with zero net magnetization as exhibited in figure 2. The presence of monovacancy (V_{C1}) does not induce the ferromagnetic ordering due to spin compensation, the local spin-density distributes symmetrically around the vacancy, resulting in no net magnetic moment, but slightly decreases the interlayer spacing to 3.46 Å. The stabilization of carbon vacancy with hydrogen atom introduces significant magnetization in the structure having magnetic moment of 2.87 μ_B. It has also been noticed that di-vacancy configurations ($V_{C1\text{-}C2}$ and $V_{C1\text{-}C3}$) exhibit significant magnetic ordering having magnetic moment value of 2.13 μ_B and 2.10 μ_B, respectively with degree of spin-polarization up to 56.65%. The significant change in magnetic moment with respect to the vacancy concentration and vacancy distance, exhibits the influence of defect spacing and the vacancy interaction on emerging magnetic ordering. This corroborates well with the experimental findings of increased saturation magnetization and enhanced I_D/I_G ratios at optimal fluence of 35 pc μm^{-2} for HPG-2 sample, where optimal carbon vacancies and hydrogen incorporation coexist. It has also been observed that hydrogen at interstitial (H_{int}) site induces a small magnetic moment of 0.02 μ_B, but significantly enhances the interlayer spacing up to 3.60 Å. The significant interlayer expansion corroborates well with the experimental XRD and Raman analysis. This suggests that interlayer hydrogen plays an indirect but crucial role in regulating the structural and electronic environment that promotes magnetism.

5. Conclusions

This work presents the formation of room-temperature ferromagnetic ordering in Highly Oriented Pyrolytic Graphite (HOPG) upon 1.2 MeV proton (H^+) ion implantation, with an optimum fluence of 35 pc μm^{-2}. The structural changes such as lattice distortion, defect formation and interlayer expansion have been revealed by X-ray diffraction (XRD) and Raman spectroscopy. Vibrating Sample Magnetometry (VSM) reveals that the proton fluence of 35 pc μm^{-2}, enhances the saturation magnetization up to a value of 0.89 emu g^{-1} in the HPG-2 sample. The proton ion stopping range, ion distribution and respective vacancy formation has been estimated utilizing SRIM simulation. It has been noticed that proton ion creates maximum carbon vacancy at the end of projection range. The contribution of individual defects, such as mono-vacancy, di-vacancy, Hydrogen incorporation (at interstitial and at carbon vacancy site) in HOPG, towards magnetization, has also been elucidated utilizing spin-polarized DFT simulation. DFT simulations suggest that hydrogen incorporation at carbon vacancies and interstitial sites, and the interaction of di-vacancies, are crucial for the onset of magnetic ordering in carbon-based system. A single hydrogen filled at mono-vacancy site enhances the magnetic moment up to 2.87 μ_B and supports the consideration of proton ion for implantation and defect creation. Our findings reveals the mechanism behind defect induced magnetism in HOPG, utilizing experimental and theoretical approaches and provide a pathway for developing lightweight magnetic materials for future spintronic and carbon-based device application with a better theoretical understanding of useful defects leading to ferromagnetism at room temperature.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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